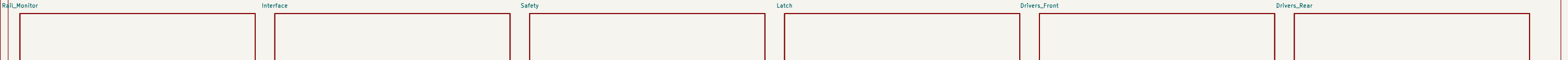
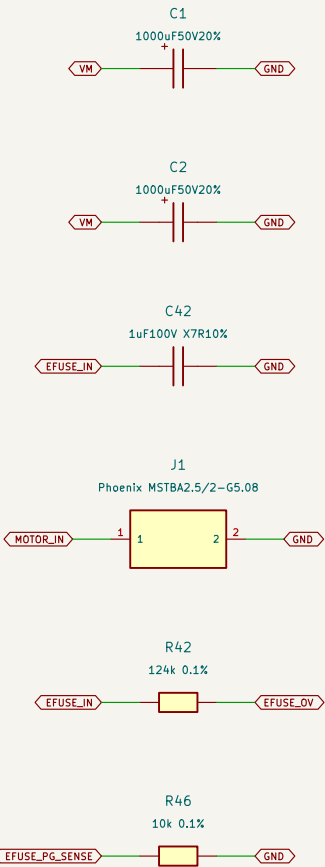
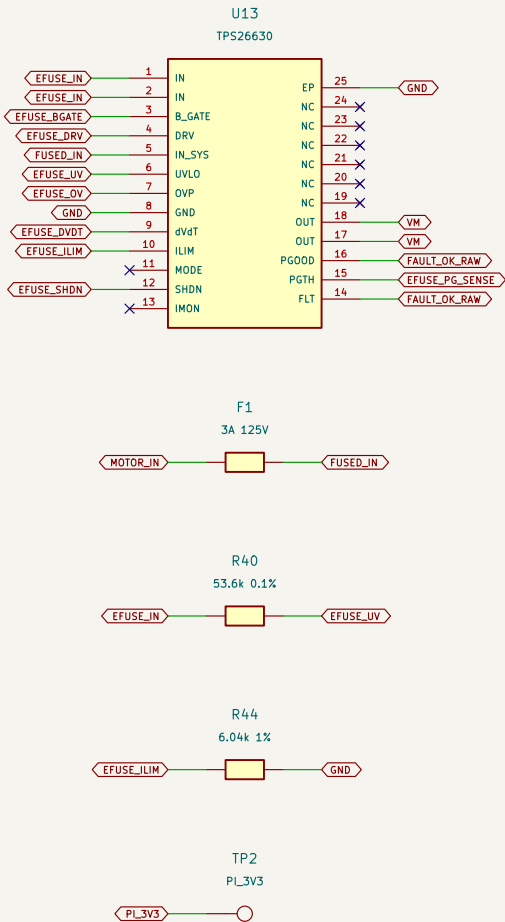
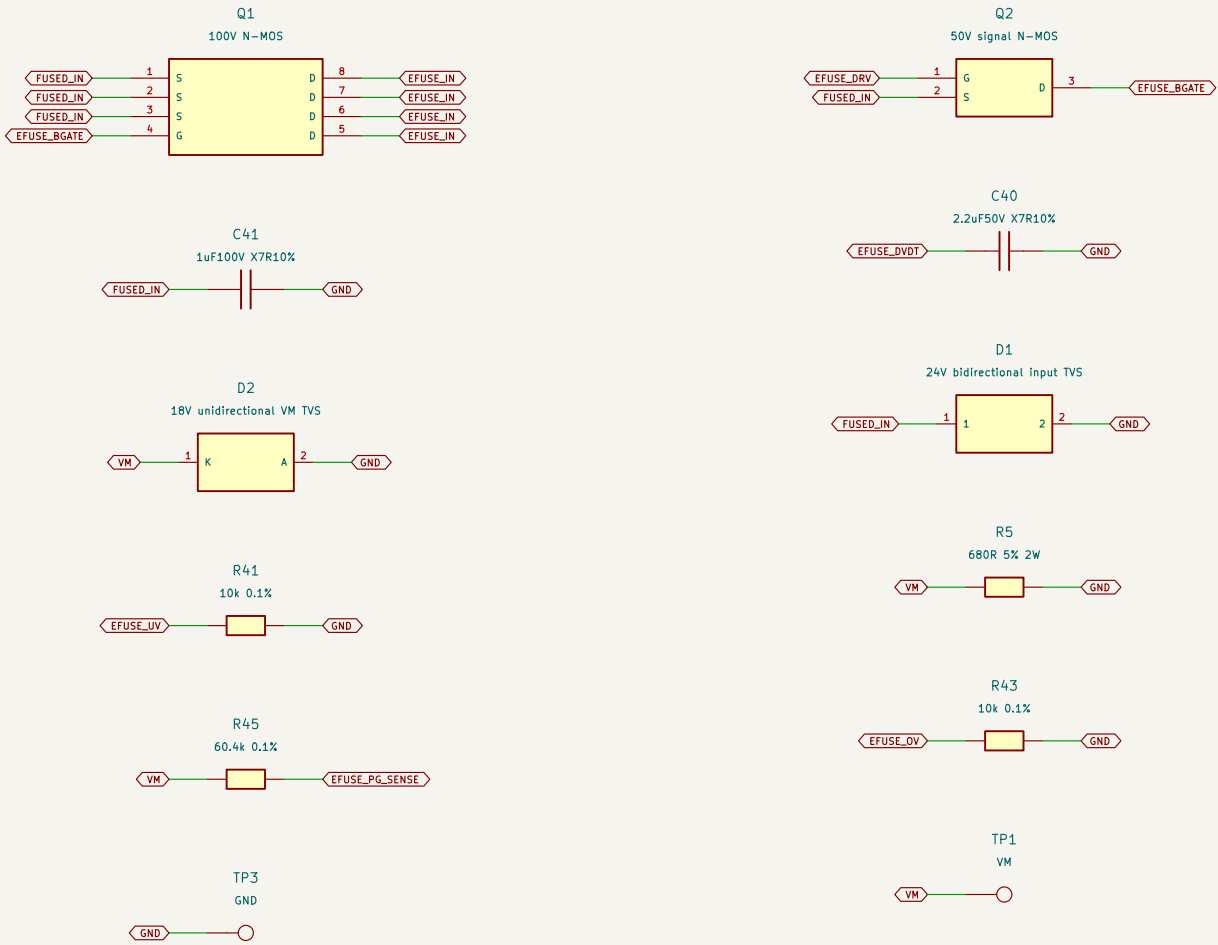


12 V class input protection / controlled feed

PROVISIONAL 9–15 V source envelope; actual pack chemistry / maximum / topology UNCONFIRMED.
eFuse limits aggregate input current; driver choppers limit individual motors. Regeneration <=0.1 J/event, <=0.1 Hz.

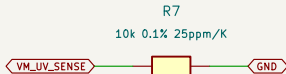
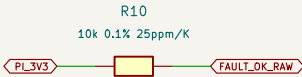
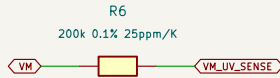
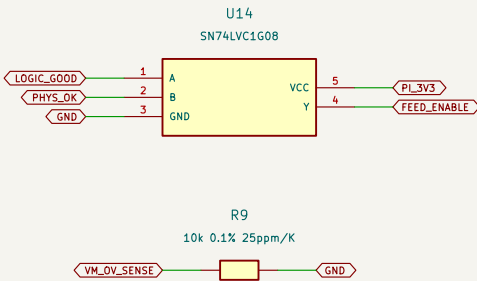
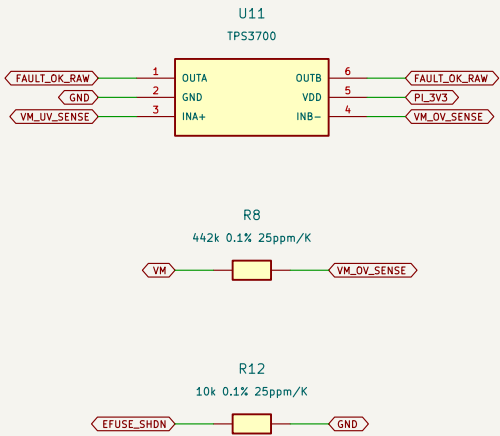


File: rail_monitor.kicad_sch File: interface.kicad_sch File: supervision.kicad_sch File: enable_latch.kicad_sch File: motor_front.kicad_sch File: motor_rear.kicad_sch

Rail / fault qualification and physical feed control

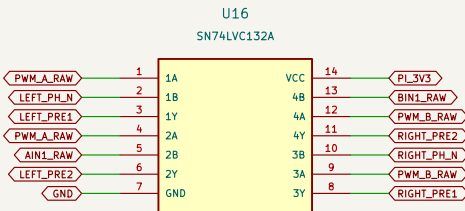
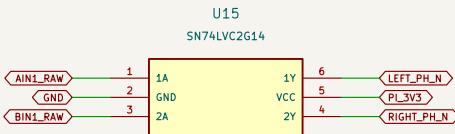
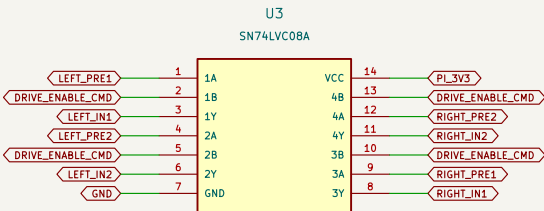
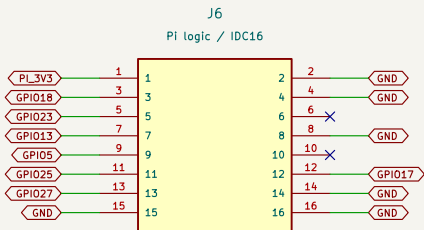
SW1 INHIBIT drops nSLEEP permission and opens the eFuse feed; stored VM energy remains.

FAULT_OK_RAW is wired-AND; VM monitor, eFuse PGOOD/FLT and four driver nFAULT outputs. Any fault clears the arm latch.



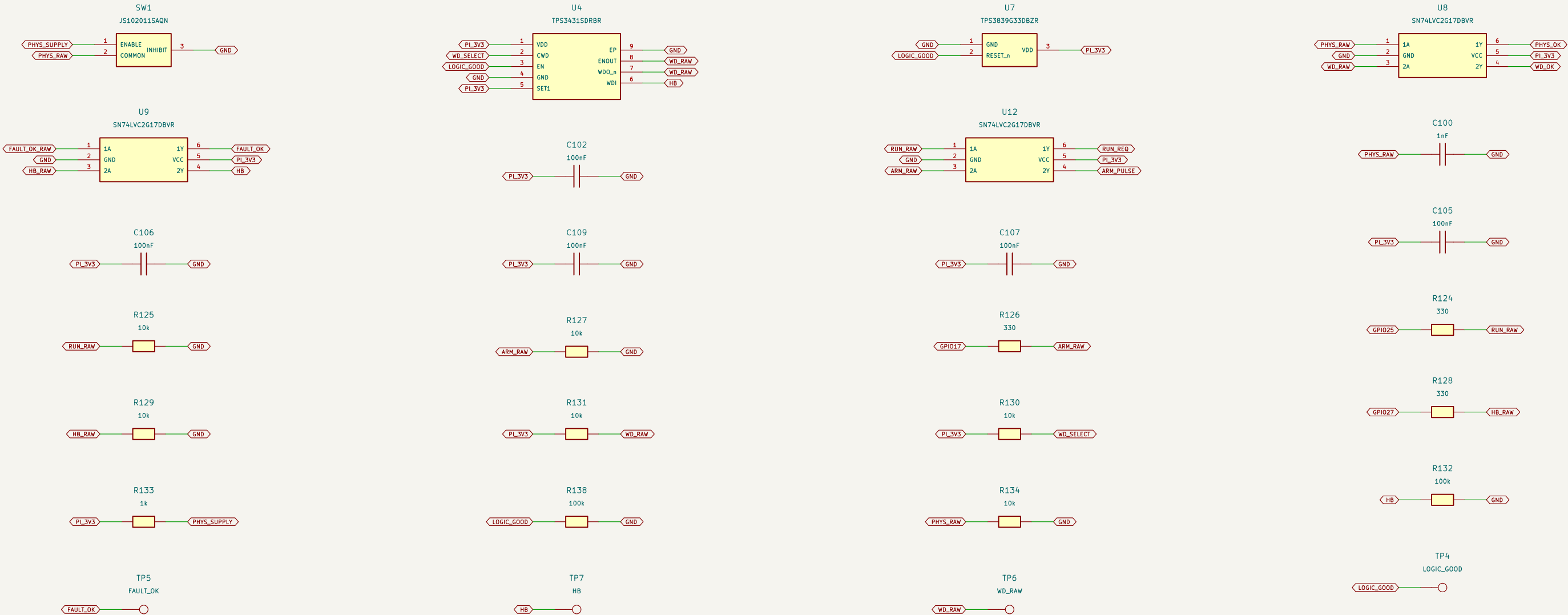
Pi interface / paired PWM and DIR commands

Custom 16-way cable, NOT the Pi 40-pin header. Pins 6/10 intentionally NC; new harness / backend required.
Schmitt NAND decode + final ARM-AND: armed PWM-off=brake; disarmed=coast. One separate output pair per motor.



Watchdog / manual disable / input conditioning

FEED_ENABLE wakes drivers before ARM; normal wake nFAULT cannot create a sleep/fault feedback loop.
GPIO17 ARM; GPIO27 heartbeat; GPIO25 RUN. Pi commands are PWM/DIR; decoded driver mode is PWM. Software remains mock-only.



PROVISIONAL – NOT VERIFIED FOR THE EXISTING ROBOT
NOT BUILT / NOT PHYSICALLY TESTED

Design only / pre-fabrication

Sheet: /Safety/
File: supervision.kicad_sch

Title: AI Robot Rev A / Safety

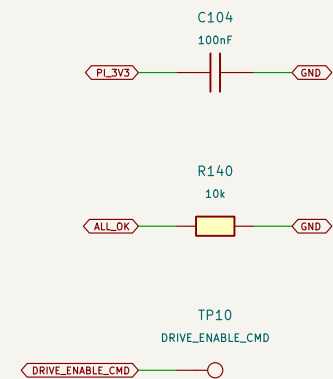
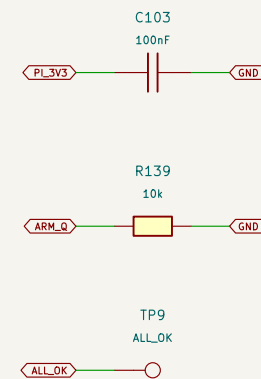
Size: A3 Date: 2026-09-28
KiCad E.D.A. 10.0.6

Rev: A-12V-DRAFT
Id: 4/7

Deliberate arm latch and hardware sleep permission

Loss of ALL_OK asynchronously clears ARM_Q. Recovery requires a fresh ARM edge after valid conditions.

DRIVE_ENABLE_CMD gates both inputs of every bridge; nSLEEP follows physical FEED_ENABLE, independently of ARM/fault.



PROVISIONAL – NOT VERIFIED FOR THE EXISTING ROBOT
NOT BUILT / NOT PHYSICALLY TESTED

Design only / pre-fabrication

Sheet: /Latch/
File: enable_latch.kicad_sch

Title: AI Robot Rev A / Latch

Size: A3	Date: 2026-09-28
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Size: A5	
KiCad E.D.A. 10.0.6	

Date: 2026-09-28

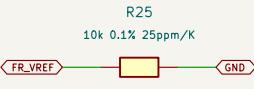
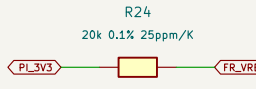
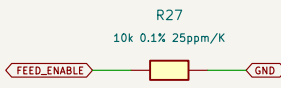
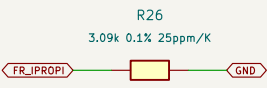
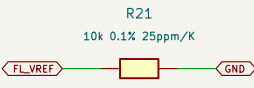
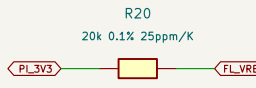
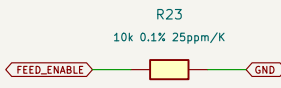
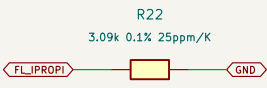
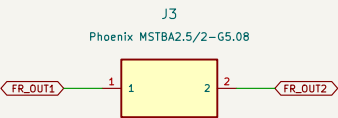
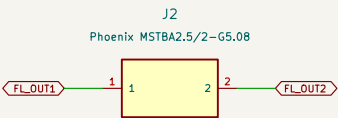
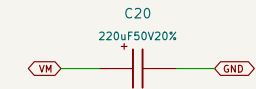
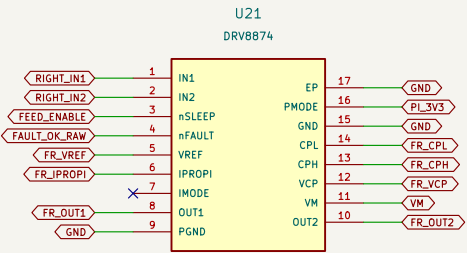
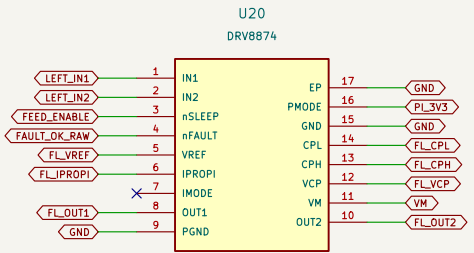
Rev: A-12V-DRAFT

Id: 5/7

Front motors / independent regulated H bridges

U20 FL / U21 FR: one DRV8874 per motor, native current regulation approx 0.79 A; IMODE floating.

Candidate motor identity remains unconfirmed. Startup torque and motor stall heating are NOT qualified.



PROVISIONAL – NOT VERIFIED FOR THE EXISTING ROBOT
NOT BUILT / NOT PHYSICALLY TESTED

Design only / pre-fabrication

Sheet: /Drivers_Front/
File: motor_front.kicad_sch

Title: **AI Robot Rev A / Drivers_Front**

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-28

Rev: **A-12V-DRAFT**
Id: 6/7

Rear motors / independent regulated H bridges

U22 BL / U23 BR: same paired side commands, separate outputs, current sensing and regulation per motor.
37 V operating driver limit; reviewed VM transient ceiling 32 V. No encoder interface or physical validation claimed.

